



CHAPTER

6

Redox Reactions

Sr. No.	Concept	Formulas	Other information
1.	Oxidation & Reduction		Oxidation = Loss of electron Reduction = Gain of electron
2.	Oxidation Number		
	(a) Free State (b) Monoatomic ions (c) Oxygen (i) Peroxides (ii) Superoxides (iii) Oxygen difluoride (iv) Dioxygen difluoride (d) Hydrogen (e) Fluoride	O. No. of element in Free State = 0 O. No. of monoatomic ions = Charge of ions O. No. of oxygen = -2 O. No. of Peroxides = -1 O. No. of Superoxides = $-\frac{1}{2}$ O. No. of Oxygen difluoride = +2 O. No. of Dioxygen difluoride = +1 O. No. of Hydrogen = +1 O. No. of Fluoride = -1	O. No. = Oxidation Number
3.	Equivalent weight		
	(a) Equivalent weight of a salt (b) Equivalent weight of acids (c) Equivalent weight of bases	Equivalent weight of a salt (E_w) $= \frac{\text{Molar mass}}{\text{Net positive (or negative) valency}}$ Equivalent weight of acids = $\frac{\text{Molar mass}}{\text{Basicity}}$ Equivalent weight of bases = $\frac{\text{Molar mass}}{\text{Acidity}}$	Basicity is the number of replaceable H^+ ions. Acidity is the number of replaceable OH^- ions.
4.	No. of gram- equivalents (Eq)	$\text{No. of gram equivalents (Eq)} = \frac{\text{Weight of compound}}{\text{Equivalent weight}}$	
5.	Mole equivalent relationship	$n = \frac{w}{M_w} \text{ and } Eq = \frac{w}{\text{Equivalent weight}}$ $\frac{Eq}{n} = \frac{M_w}{\text{Equivalent weight}} = n\text{-factor}$	n = No. of moles of element w = mass of element M_w = Molar mass of element

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6.	Normality(N) & Molarity(M) relationship	$N = \frac{Eq}{V} \text{ and } M = \frac{n}{V}$ $\frac{N}{M} = \frac{Eq}{n} = \frac{M_w}{E_w} = n\text{-factor}$	n = No. of moles of element V = Volume M _w = Molecular weight E _w = Equivalent weight
7.	Equivalent weight of O.A. and R.A.	Equivalent weight of O.A. or R.A. $= \frac{\text{Molar mass}}{\text{Change in O.N. per mole}}$	O.A. = Oxidising agent R.A. = Reducing agent O.N. = Oxidation number
8.	Acid-Base Titration	Gram equivalent of acid = Gram equivalent of base	
9.	Redox Titration	Gram equivalent of O.A. = Gram equivalent of R.A.	O.A. = Oxidising agent R.A. = Reducing agent

